Document Information	
Filename	Flashing Instruction TCU for SDK Release 3.1.6.1.1 and later
Status	Draft
Revision Date	March 2, 2022
Version	1.00
Document ID	NA

Standard Proprietary Terms

THE REPRODUCTION, DISTRIBUTION AND UTILIZATION OF THIS DOCUMENT AS WELL AS THE COMMUNICATION OF ITS CONTENTS TO OTHERS WITHOUT EXPRESS AUTHORIZATION IS PROHIBITED. OFFENDERS WILL BE HELD LIABLE FOR THE PAYMENT OF DAMAGES. ALL RIGHTS RESERVED IN THE EVENT OF THE GRANT OF A PATENT, UTILITY MODEL OR DESIGN.

© Continental Automotive GmbH 2020
Confidential and Proprietary

Table of Contents

1 Introduction 3
1.1 Install the USB Drivers: 3
1.2 Install the QPST Tool 3
1.3 Backup Calibration Data 4
1.3.1 Start QPST App. 4
1.3.2 Backup file naming convention (recommended) 5
Note for P1 HW Calibration Backup 6
1.4 Flash the image using QFIL. 7
1.4.1 Updating the Chestnut board 7
1.4.2 Updating the Harman TCU board 7
1.5 Update the QCN file. 9
1.5.1 P1 HW Calibration Restoration: 11
1.6 Install Persistency database 11
1.7 Install the modem configuration 11
1.7.1 Pushing Modem Configurations to device. 11
1.7.2 Checking and Changing Modem profile: 11
Troubleshooting 13

© Continental Automotive GmbH 2020
Confidential and Proprietary

1 Introduction

Starting the SDK release Audi SDK tp_sdk_otp-sa515m-thick-3.1.3.10.1_pkg, NAD requires some changes in the previous flashing instruction because new software requires updated QCN.

Here are the instructions and steps.

1.1 Install the USB Drivers:

a) Install the NAD USB Driver. **qud.win.1.1_installer_10061.1.zip**

b) Unzip the folder and double click on the installer.

c) Select the WWAN-DHCP and click on next until installation finishes

1.2 Install the QPST Tool

Unzip and double click on installer and continue until installation finishes.

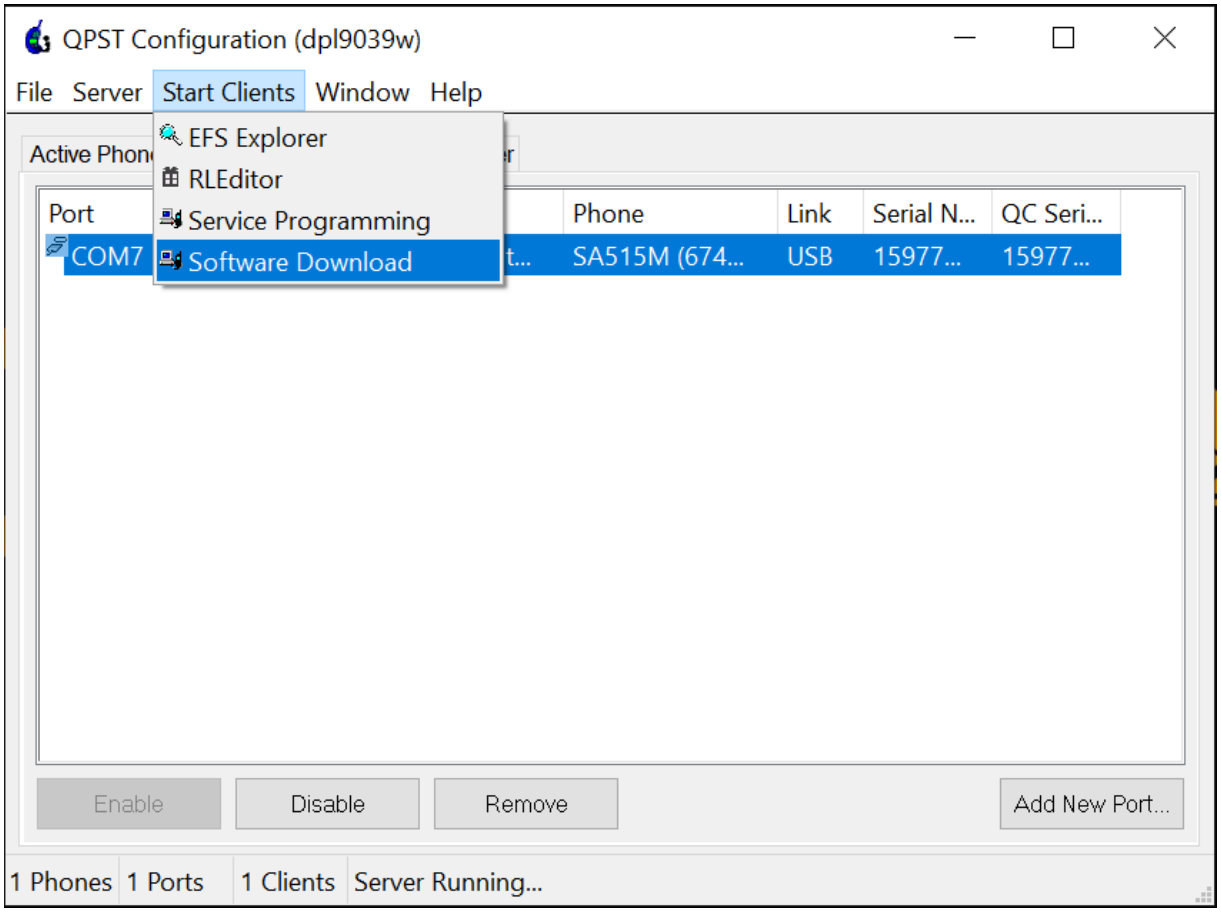
qpst.win.2.7_installer_00480.14.zip

Backup the XQCN data from device.
Always do this before carrying out any other operation on the device. Keep this file in safe location for possible future recovery of device if needed!

© Continental Automotive GmbH 2020
Confidential and Proprietary

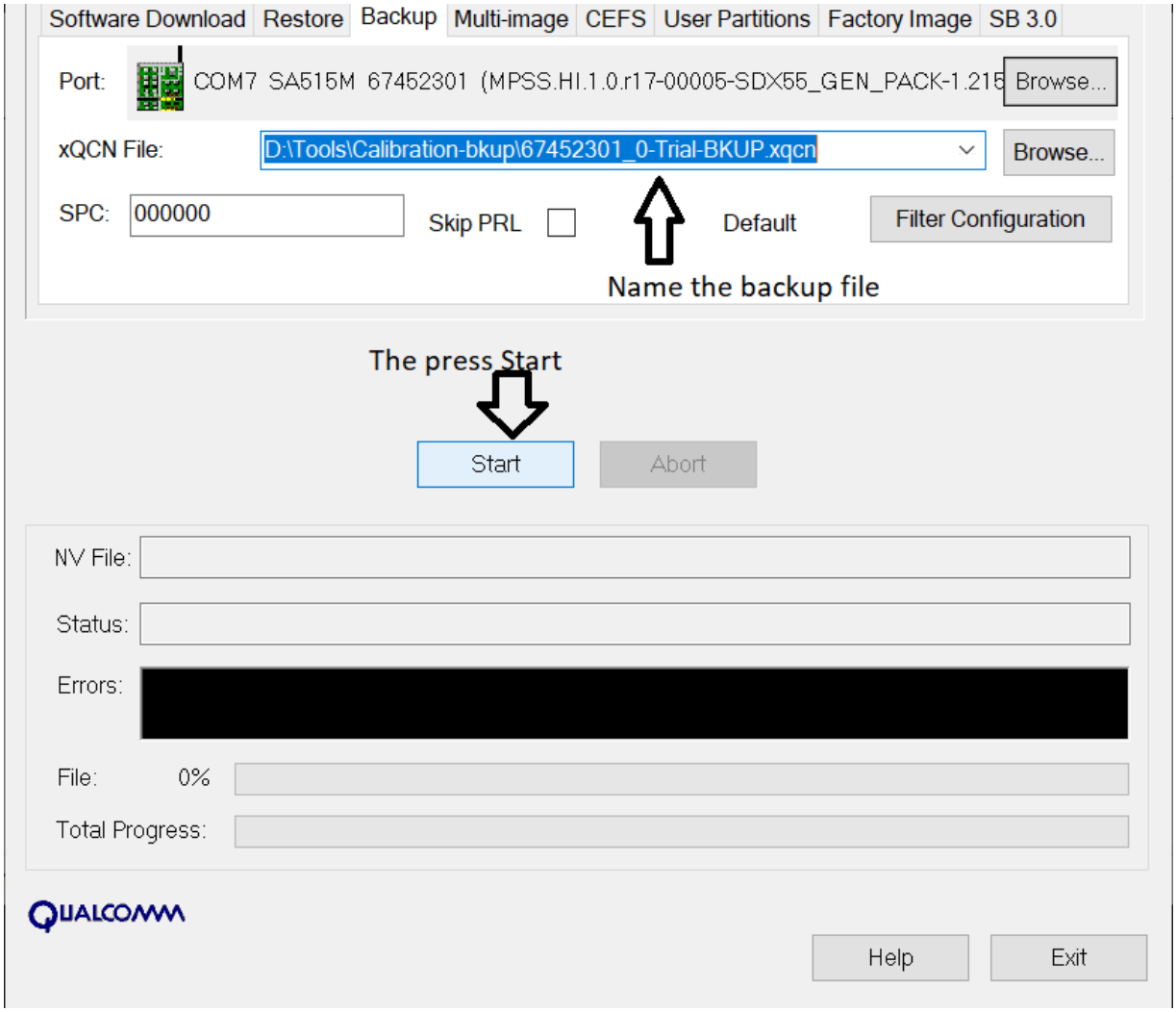
1.3 Backup Calibration Data

1.3.1 Start QPST App.



© Continental Automotive GmbH 2020
Confidential and Proprietary

To backup calibration data or XQCN (QCN file = Calibration data).



1.3.2 Backup file naming convention (recommended)

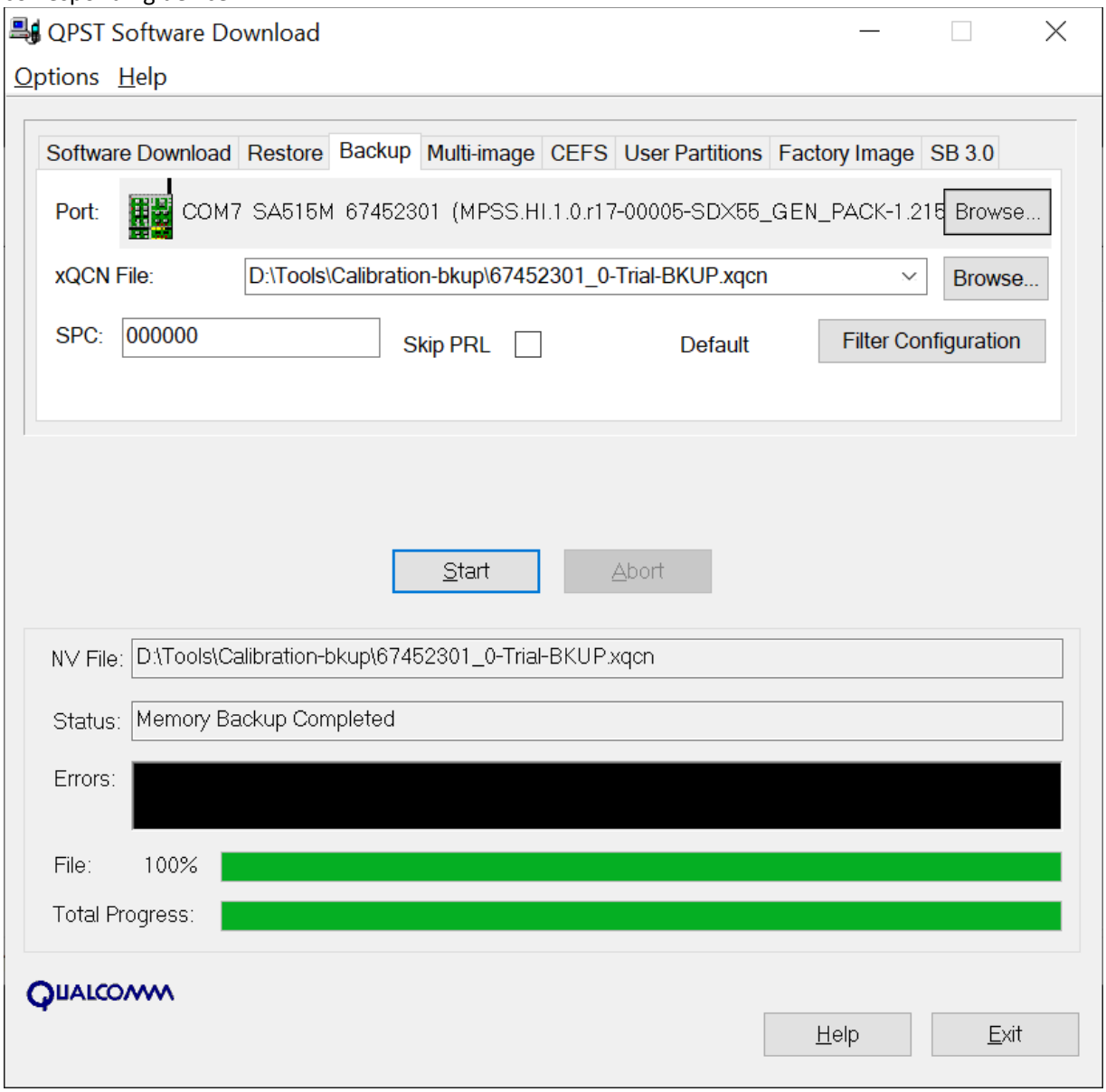
It always useful to be able to match the device with the file from name of file. For this reason following naming convention is strongly recommended.

<Device Type>-<HW Ver>_<Serial number>_<IMEI>
<Harman-B1TCU>_<P05-3>_<SN42>_<004401810788154>.xqcn

So actual file in this example would be;
Harman-B1-TCU_P05-3_SN42_004401810788154.xqcn

© Continental Automotive GmbH 2020
Confidential and Proprietary

When restoring the XQCN file using QPST App it should be easy to ensure that correct file is restored to the corresponding device.



Backup completed.

Note for P1 HW Calibration Backup

In case of P1 sa515m targets, calibration backup is NOT required as P1 sa515m are not calibrated at the time of manufacture., hence a special xQCN known as 'golden' xQCN has been created for P1 sa515m targets (P1.1.1 and P1.2.1).
P1 sa515m golden xQCN (QCN_GOLD_P1-1.1_590.xqcn) for Audi/ is embedded here (P1 USE ONLY).

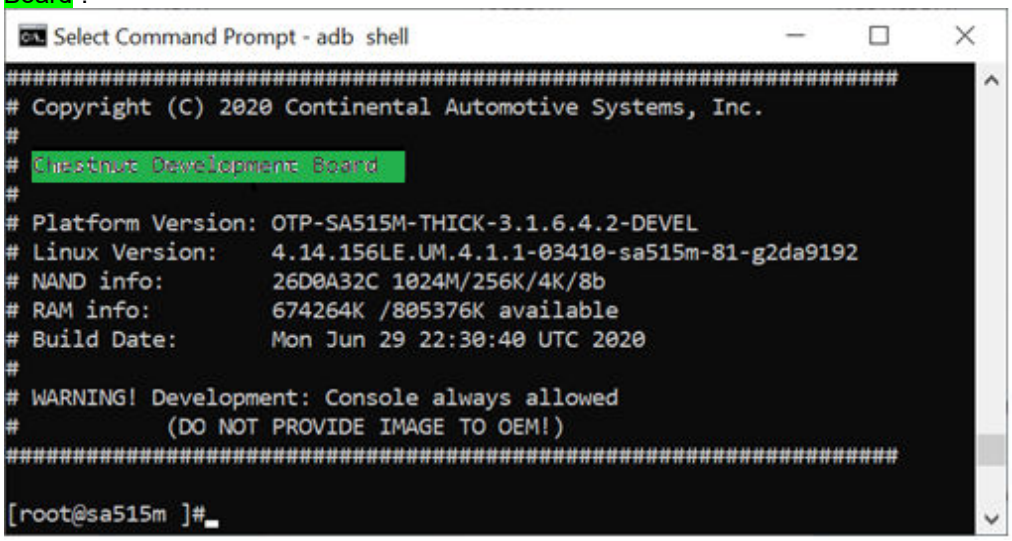
© Continental Automotive GmbH 2020
Confidential and Proprietary

1.4 Flash the image using QFIL.

a) Before flashing the image, please ensure that correct SBL file is placed in the 4K folder.

1.4.1 Updating the Chestnut board

updating the Chestnut board, then default SBL as delivered in Conti Build is correct one to use and QFIL will use this. After flashing Chestnut Board splash screen should be like this "Chestnut Development Board".



1.4.2 Updating the Harman TCU board

b) Please delete the existing (default) sbl1.mbn, then copy the sbl1-audi-conmod.mbn and rename to **sbl1.mbn**. QFIL will pick up this file only.

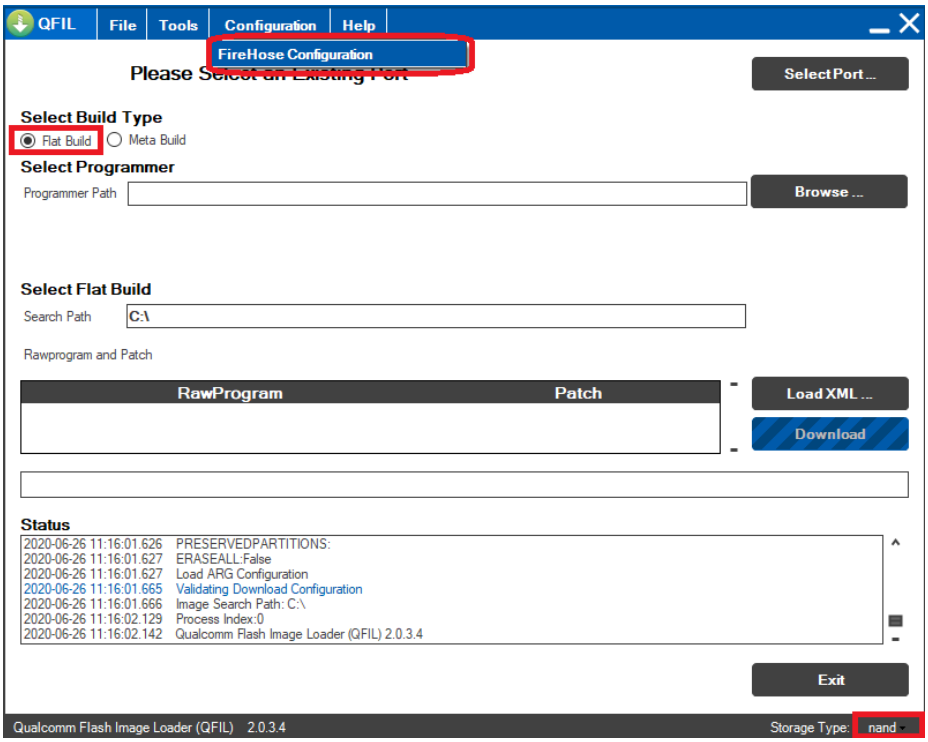
After flashing Harman TCU, the splash screen should be like this "Audi ConMod Bx".

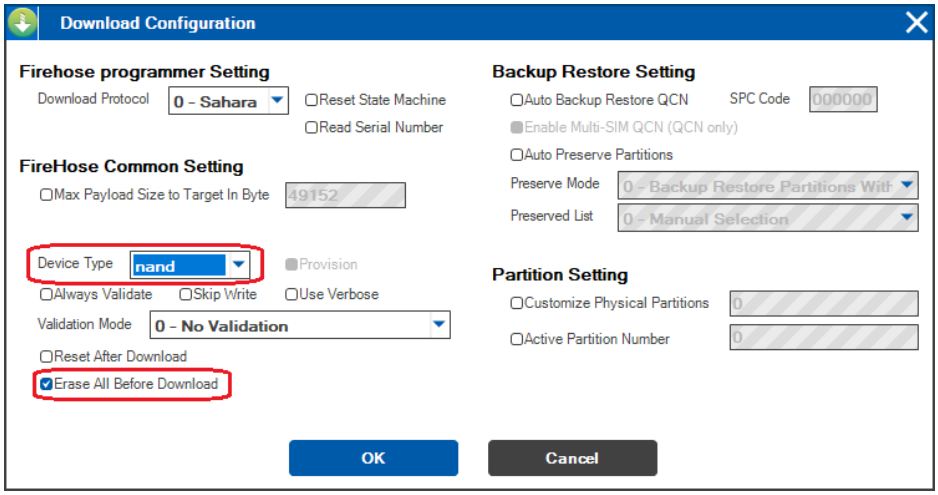
```
##########################################################################
# Copyright (C) 2020 Continental Automotive Systems, Inc.
#
# Audi ConMod B1
#
# Platform Version: OTP-SA515M-THICK-3.1.6.4.2-DEVEL
# Linux Version:    4.14.151LE.UM.4.1.1-03210-sa515m-75-ge16b38b
# NAND info:        26D0A32C 1024M/256K/4K/8b
# RAM info:         596624K /805376K available
# Build Date:       Wed Mar 11 23:59:53 UTC 2020
#
# WARNING! Development: Console always allowed
#          (DO NOT PROVIDE IMAGE TO OEM!)
##########################################################################
```

It is critical to ensure the correct SBL file use for the corresponfin device.

c) Open the QFIL tool and select Configuration → FireHose Configuration → Select NAND and select "Erase all before downloand" as shown below.

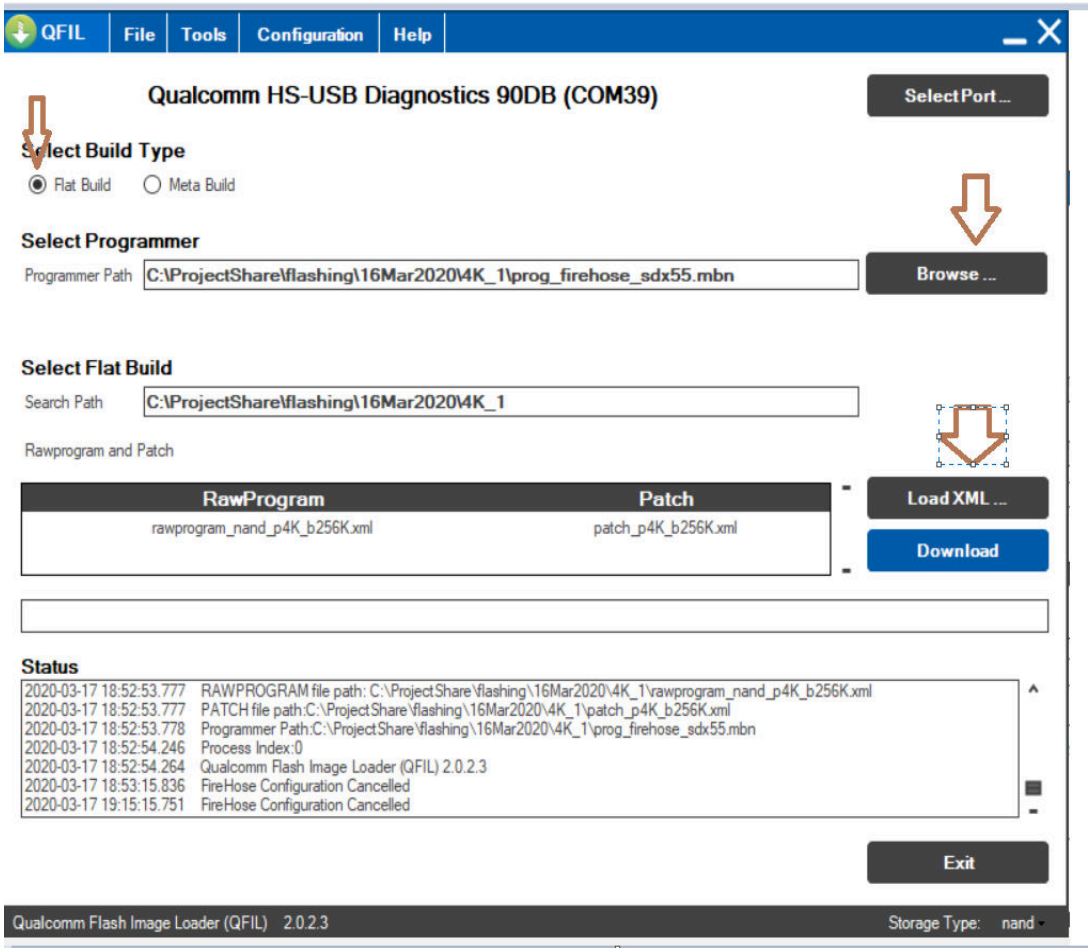
© Continental Automotive GmbH 2020
Confidential and Proprietary





d) Select build type “Flat”, Browse the path and selec the prog_firehose_sdx55.mbn from 4K folder and then Load XML. Load XM will automatically select the file.

© Continental Automotive GmbH 2020
Confidential and Proprietary



e) Click on download and wait for device to be flashed.
f) Once flashing is succesfull, power cycle (turn power off and back on) the unit.

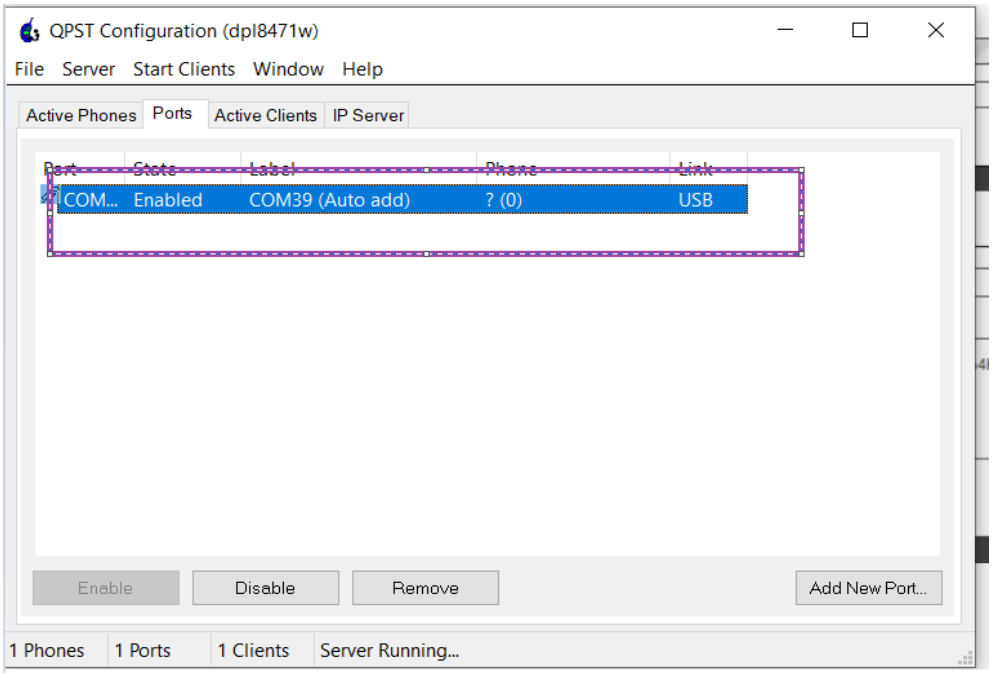
g) **Hint:** *If the download fails with the message "failed to enter EDL mode", in adb shell command prompt enter command (within a few seconds! of the error message)*
> sys_reboot edl

This should get the Download started.
Perform full Power cycle (disconnect power completely) after download is completed.

1.5 Update the QCN file.

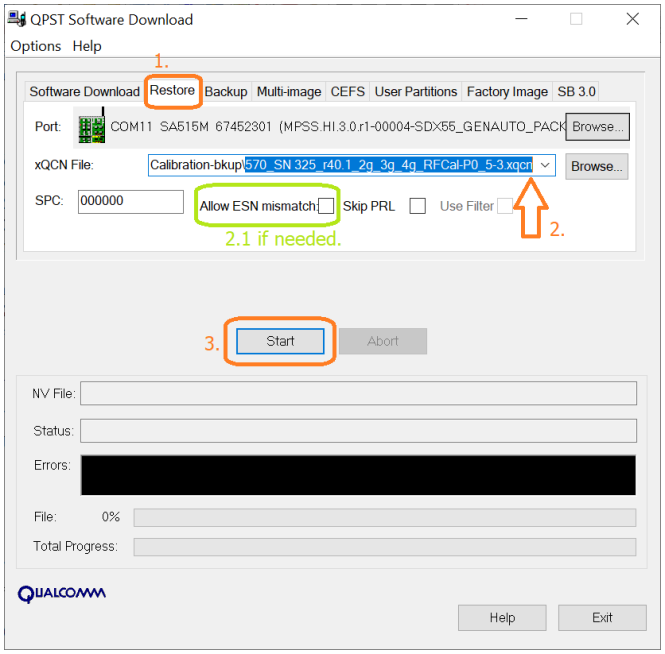
a) Open the QPST tool and make sure that COM port is detected in the tool

© Continental Automotive GmbH 2020
Confidential and Proprietary



b) Go the the Start Clients tab and select Software download.
c) Go to the restore tab and select the new QCN-file from browse and click Start.

Use the same QCN-file which you have already backup from the device before image flashing in section 1.3



In case you see an error in the “Errors” dialogue, tick the (2.1) box “Allow ECN Mismatch”, the click Start again.

© Continental Automotive GmbH 2020
Confidential and Proprietary

1.5.1 P1 HW Calibration Restoration:

'golden' xQCN specified in note for P1 sa515m hardware can be restored as mentioned in section 1.5

1.6 Install Persistency database

This is a 3 step process. Execute **from Window CMD** via commands;

I. First Remove the old persistency:
- `adb shell rm -rf /data/persistency`

II. Second: Push the new persistency to device.
- `adb push tp-pers-install.tar.gz /tmp`

  Response should be: 1137 KB/s (5813 bytes in 0.004s)

III. Third Install the new persistency
- `adb shell pers-resource-installer /tmp/tp-pers-install.tar.gz`

  Response should be: Successful installation

a) Perform Reboot

1.7 Install the modem configuration

Pushing of the modem is needed. However the profile selection/override should be done for testing purposes ONLY. By default system will use the SIM ICCID to select the correct profile automatically. No manual intervention necessary.

1.7.1 Pushing Modem Configurations to device.

a) tp_modem_config_list.tar should be present inside the 4K Build folder. Push the file to target through this adb command below. After the PUSH system will perform auto reboot!.
- ***adb push tp_modem_config_list.tar.gz /tmp***
  adb shell mv /tmp/tp_modem_config_list.tar.gz /tmp/modem_config
  After push is completed system should auto reboot, in case it does not, perform reboot manually. Once reboot is completed check for the modem cofig list as follows:

1.7.2 Checking and Changing Modem profile:

Modem Config List:
At the adb shell prompt issue the command below;
- ***mcm-test -i***

You should see the screen shot below. Here you can the state of the profile, which are Active and which are Inactive. You can also modify and select the profile of interest, as shown below.

© Continental Automotive GmbH 2020
Confidential and Proprietary

Select Command Prompt - adb shell

```
0) Exit
1) Get Config list
2) Set override modem configuration.
3) Get override modem configuration state.
4) Disable Override.
5) Delete configurations
1

1. mcfg_sw_ROW_Commercial.mbn [Version : a010809] -> Active
2. mcfg_hw_FE515_LE_SS.mbn [Version : a018004] -> Active
3. mcfg_sw_Vodafone_VoLTE_Turkey.mbn [Version : a0104c6] -> Inactive
4. mcfg_sw_Vodafone_VoLTE_Germany.mbn [Version : a010449] -> Inactive
5. mcfg_sw_Verizon_CDMAless.mbn [Version : a010126] -> Inactive
6. mcfg_sw_KT_Commercial_KT_LTE.mbn [Version : a01280b] -> Inactive
7. mcfg_sw_DT_VoLTE_Commercial.mbn [Version : a011f1f] -> Inactive
8. mcfg_sw_DCM_Commercial.mbn [Version : a010d0d] -> Inactive
9. mcfg_sw_ATT_VoLTE.mbn [Version : a010335] -> Inactive
10. mcfg_sw_AMX_Commercial_MX.mbn [Version : a011600] -> Inactive
Total: 10
Enter config items to activate (comma separated):
Example:1,2
2,4
2,4
Activate-> mcfg_hw_FE515_LE_SS.mbn
Activate-> mcfg_sw_Vodafone_VoLTE_Germany.mbn
Successfully set override.
0) Exit
1) Get Config list
2) Set override modem configuration.
3) Get override modem configuration state.
4) Disable Override.
5) Delete configurations
1
1
1. mcfg_sw_Vodafone_VoLTE_Germany.mbn [Version : a010449] -> Active
2. mcfg_hw_FE515_LE_SS.mbn [Version : a018004] -> Active
3. mcfg_sw_ROW_Commercial.mbn [Version : a010809] -> Inactive
4. mcfg_sw_Vodafone_VoLTE_Turkey.mbn [Version : a0104c6] -> Inactive
5. mcfg_sw_Verizon_CDMAless.mbn [Version : a010126] -> Inactive
6. mcfg_sw_KT_Commercial_KT_LTE.mbn [Version : a01280b] -> Inactive
```


© Continental Automotive GmbH 2020
Confidential and Proprietary

Troubleshooting

What if the board is stuck in one of the boot stages and Continental fastboot client is not working. How can I recover the board?

There are multiple boot stages inside the NAD. Pleaes check the serial console log from the NAD and if you see some logs from UEFI boot stages. Logs like as shown below. It means that target is stuck in UEFI mode and it can still be recovered.

```
[3.16 16:08:36.6]UEFI Start     [  817] SEC

[3.16 16:08:38.1]Error Reading FRP partition: Not Found
[3.16 16:08:38.1]Fastboot: Processing commands
[3.16 16:08:38.1]Failed to get the graphics output protocol.
[3.16 16:08:38.1]Failed to get the graphics output protocol.
[3.16 16:08:38.1]Failed to get width or height
[3.16 16:08:38.1]Failed to get the graphics output protocol.
[3.16 16:08:38.1]Failed to get the graphics output protocol.
```

Please use the fastboot client attached here to flash the unit and board can still be recovered with the special client.
\\auto.contiwan.com\cas\Loc\dep1\didt6639\TP_Transfer\Bhanu\ConMod\fastboot_uefi

Also to debug such issue is recommended that back up of raw NAND is taken before the unit is reflashed. Standard fastboot command to dump the image and flash would work and board can be recovered.

If board is stuck at earlier boot stages before UEFI

Restart the board and if board is always stuck here then only way to recover the board is with EDL. Regarding EDL pin on the TCU please contact the TCU Supplier. Once EDL pin is set, QFIL should be used to flash the software.

What if ethernet is not working and software looks to be broken, what can be done as initial verification that flashing of software is correct.

One of the common problem is not selecting the correct SBL during flashing. Please ensure that correct SBL has been flashed and basic verification can be done by looking that software banner and looking through the software version along.

© Continental Automotive GmbH 2020
Confidential and Proprietary

```
#######################################################################
# Copyright (C) 2020 Continental Automotive Systems, Inc.
#
# Audi ConMod B1
#
# Platform Version: OTP-SA515M-THICK-3.1.3.10.1-DEVEL
# Linux Version:    4.14.151LE.UM.4.1.1-03210-sa515m-75-ge16b38b
# NAND info:        26D0A32C 1024M/256K/4K/8b
# RAM info:         596624K /805376K available
# Build Date:       Wed Mar 11 23:59:53 UTC 2020
#
# WARNING! Development: Console always allowed
#          (DO NOT PROVIDE IMAGE TO OEM!)
#######################################################################
```

Checking the content of modem Config file:
This can be at adb shell with command >mcm-test -i
Folow the steps indicated. The list content (in this example 6, should be exactly same the file that PUSHed to the device.

```
Command Prompt - adb shell
C:\windows\system32>adb shell

#######################################################################
# Copyright (C) 2020 Continental Automotive Systems, Inc.
#
# Chestnut Development Board
#
# Platform Version: OTP-SA515M-THICK-3.1.4.4-DEVEL
# Linux Version:    4.14.151LE.UM.4.1.1-03210-sa515m-77-g2f40494
# NAND info:        26D0A32C 1024M/256K/4K/8b
# RAM info:         588736K /805376K available
# Build Date:       Tue Apr  7 14:42:22 UTC 2020
#
# WARNING! Development: Console always allowed
#          (DO NOT PROVIDE IMAGE TO OEM!)
#######################################################################

[root@sa515m ]#mcm-test -i
mcm-test -i
MCM-TEST started
0) Exit
1) Get Config list
2) Set override modem configuration.
3) Get override modem configuration state.
4) Disable Override.
5) Delete configurations
1
1
1. mcfg_sw_ROW_Commercial.mbn [Version : a010809] -> Active
2. mcfg_hw_FE515_LE_SS.mbn [Version : a018002] -> Active
3. mcfg_sw_Vodafone_VoLTE_Germany.mbn [Version : a010449] -> Inactive
4. mcfg_sw_Verizon_CDMAless.mbn [Version : a010126] -> Inactive
5. mcfg_sw_DT_VoLTE_Commercial.mbn [Version : a011f1f] -> Inactive
6. mcfg_sw_ATT_VoLTE.mbn [Version : a010335] -> Inactive
Total: 6
```


© Continental Automotive GmbH 2020
Confidential and Proprietary

The Modem configuration is not correct, how do update to correct content.

Start ADB Shell, execute mcm-test -I, and follow instruction in screen shot below,

```
Select Command Prompt - adb shell
[root@sa515m modem_config]#mcm-test -i
mcm-test -i
MCM-TEST started
0) Exit
1) Get Config list
2) Set override modem configuration.
3) Get override modem configuration state.
4) Disable Override.
5) Delete configurations
1
1
1. mcfg_sw_ROW_Commercial.mbn [Version : a010809] -> Active
2. mcfg_hw_FE515_LE_SS.mbn [Version : a018002] -> Active
3. mcfg_sw_Vodafone_VoLTE_Germany.mbn [Version : a010449] -> Inactive
4. mcfg_sw_Verizon_CDMAless.mbn [Version : a010126] -> Inactive
5. mcfg_sw_DT_VoLTE_Commercial.mbn [Version : a011f1f] -> Inactive
6. mcfg_sw_ATT_VoLTE.mbn [Version : a010335] -> Inactive
Total: 6
0) Exit
1) Get Config list
2) Set override modem configuration.
3) Get override modem configuration state.
4) Disable Override.
5) Delete configurations
5
5
1. mcfg_sw_ROW_Commercial.mbn [Version : a010809] -> Active
2. mcfg_hw_FE515_LE_SS.mbn [Version : a018002] -> Active
3. mcfg_sw_Vodafone_VoLTE_Germany.mbn [Version : a010449] -> Inactive
4. mcfg_sw_Verizon_CDMAless.mbn [Version : a010126] -> Inactive
5. mcfg_sw_DT_VoLTE_Commercial.mbn [Version : a011f1f] -> Inactive
6. mcfg_sw_ATT_VoLTE.mbn [Version : a010335] -> Inactive
7. All configurations
Total: 6
Enter config items to delete (comma separated):
Example:1,2
7
7
Removing configurations
mcfg_sw_ROW_Commercial.mbn -> removed from EFS and Persistency
mcfg_hw_FE515_LE_SS.mbn -> removed from EFS and Persistency
mcfg_sw_Vodafone_VoLTE_Germany.mbn -> removed from EFS and Persistency
mcfg_sw_Verizon_CDMAless.mbn -> removed from EFS and Persistency
mcfg_sw_DT_VoLTE_Commercial.mbn -> removed from EFS and Persistency
mcfg_sw_ATT_VoLTE.mbn -> removed from EFS and Persistency
No modem configurations loaded
All configurations removed
```

Then PUSH the modem config file with correct content via the adb command below
adb push tp_modem_config_list.tar.gz /tmp/modem_config

Use mcm-test -i App to check the modem config content.

© Continental Automotive GmbH 2020
Confidential and Proprietary